🔵 Studio Level 1 — Online Lesson Programme

> *A 6-week project-based programme for students aged 12 and above. Students build and deploy a real music web application — live on the internet.*

📢 Dear Parents — Why This Matters

There is a significant difference between a student who *uses* technology and one who *builds* it. Studio Level 1 is where that shift happens.

In these 6 sessions, your child will not follow a pre-written tutorial. They will plan, design, build, debug, and deploy a complete web application — a professional-quality music sampler called **"Beat Lab"** — using the same tools and workflow used by real software developers today.

This is not a toy project. When they finish, their app will have a real URL on the internet. They can send it to friends, add it to a school portfolio, or show it to a future employer. It works on phones and desktops. It plays real audio. And every part of it was built by them.

What will your child learn?

Skill	Why It Matters
React — the world's most used framework for building websites	Powers Facebook, Instagram, Airbnb, and millions of other sites
Deploying to the internet	Transforms code from "something on my laptop" into "something anyone can visit"
Audio programming	How apps load, trigger, and control sound in a browser
Professional UI design	How to use AI to generate a polished, production-quality interface
Version control with GitHub	The industry-standard tool for saving, sharing, and collaborating on code

Skill	Why It Matters
Prompt engineering	Writing precise AI instructions to build complex features quickly
Debugging	Reading error messages, identifying problems, and fixing them — a core developer skill
AI ethics	Understanding questions of authorship, sampling rights, and AI's role in creative work

What will your child walk away with?

- A **live, deployed music web app** at a real public URL
- A **GitHub repository** — the beginning of a professional portfolio
- Experience with a **real developer workflow**: build → test → fix → deploy
- The confidence to say: *"I built that."*

📋 Programme at a Glance

	Detail
Level	Studio Level 1
Duration	6 weeks · 1 session per week · 1 hour per session
Schedule	Every Saturday · **7:00–8:00pm UK time**
Format	Online (live, teacher-led)
Language	English
Group size	Maximum **8 students** — every student gets individual attention
Age	Minimum 12 years old · No upper limit
Price	**£120 per student** for the full 6-week block
What's included	All lesson materials · Teacher support · **Claude AI account provided** · No extra tool costs
What you need	Laptop or desktop · Stable internet

> 💡 **The £120 covers everything.** Your child does not need to purchase any software, subscriptions, or tools. The Claude AI account used in lessons is provided and managed by the teacher.

📅 Session-by-Session Plan

Session 1 — Setup & First Project *(1 hour)*

> **Goal:** Every student has a working development environment and sees a live React app running on their own screen by the end of the hour.

What we do	Install Node.js and npm together as a class (teacher guides each student step by step) · Create free GitHub and Vercel accounts · Scaffold a brand new React + Vite project · Install Tailwind CSS and Howler.js · Run the app in the browser and confirm everything works · Introduction to what "Beat Lab" will look like when finished
Key concepts	What is React · What is npm · What is a development environment · Why GitHub and Vercel matter
Tools	Node.js · npm · GitHub · Vercel · Terminal · *(Cursor available if students want to browse files)*
Student takeaway	"My computer is set up like a real developer's. I can see my own app running."

> 💡 **No pre-session setup needed.** Node.js installation is done together as a class at the start of Session 1. Just bring a laptop with a stable internet connection.

Session 2 — Design Your Sampler UI *(1 hour)*

> **Goal:** A professionally designed sampler interface renders on screen, and the student understands its component structure.

What we do	Introduction to React components and JSX (how React describes what you see on screen) · Use Claude Code's `frontend-design` skill to generate a bold, studio-quality UI for the sampler · Review the generated design as a class · Make personal adjustments (colours, layout, fonts) ·

	Understand how the `SamplerPad` component and the main `App` layout relate to each other
Key concepts	React components · JSX · Props · UI design principles · What makes an interface feel professional
Tools	Claude Code · React · Tailwind CSS
Student takeaway	"I have a real-looking app design and I understand why it's structured the way it is."

Session 3 — Sounds & Pads *(1 hour)*

> **Goal:** The sampler has 8 real audio samples loaded, and each pad displays correctly with its name, key, and colour.

What we do	Find 8 royalty-free audio samples · Download and rename each file cleanly · Add them to the project's asset folder · Build the `sounds.js` configuration file that defines every pad · Build the `SamplerPad` component with its glow animation and key badge
Key concepts	Royalty-free audio · File organisation · React component props · CSS animations · Configuration files
Tools	Claude Code · React
Student takeaway	"My app has 8 pads with real sounds loaded, and each one looks exactly how I want it."

Session 4 — Make It Play *(1 hour)*

> **Goal:** Every pad plays real sound when clicked or when the matching keyboard key is pressed.

What we do	Connect Howler.js to the sound configuration so each pad plays its audio file · Add keyboard shortcuts (keys A–K trigger pads 1–8) · Build the master volume slider · Track which pad is actively playing and show the visual glow state · Test every pad with clicks and key presses

Key concepts	Howler.js · React state (`useState`) · `useEffect` and event listeners · Keyboard events · Why browsers require user interaction before playing audio
Tools	Claude Code · Howler.js · React
Student takeaway	"Every pad plays a real sound. I can perform a beat using just my keyboard."

Session 5 — Polish *(1 hour)*

> **Goal:** The app feels finished, professional, and ready for public launch.

What we do	Add a "Last played" indicator below the pads · Add a keyboard guide hint for new users · Confirm pads respond to touch on mobile devices · Add a mute toggle per pad (individual pad can be silenced) · Run `npm run build` — the final build check · Fix any errors or warnings that appear · Final visual review as a group
Key concepts	UI polish and user experience · Mobile responsiveness · Build process · Reading and fixing build errors
Tools	Claude Code · React · Terminal
Student takeaway	"My app passes the build check and looks professional on both laptop and phone."

Session 6 — Launch Day 🚀 *(1 hour)*

> **Goal:** The app is live on the internet at a real public URL. The student has shared it.

What we do	Initialise a Git repository and commit all code · Push to GitHub · Connect GitHub repository to Vercel · Deploy — watch the build logs complete · Visit the live URL · Share the link with the class · Ethics discussion: who is the author of AI-assisted code? Are royalty-free samples truly free? · Brief showcase: each student demos their sampler

Key concepts	Git commit · GitHub push · Vercel deployment · Build logs · AI authorship · Intellectual property in music
Tools	Git · GitHub · Vercel · Claude Code
Student takeaway	"My app is live. I have a real URL I can share with anyone in the world."

💷 Pricing

Full 6-week Studio Level 1 block	**£120 per student**
Payment covers	All 6 sessions · All teaching materials · **Claude AI account** · Teacher support throughout
Nothing extra to pay	No software purchases · No subscriptions · No hidden costs

✅ Requirements Checklist

Before the first session, please confirm your child has the following:

☐ **Laptop or desktop computer** — tablets and phones cannot run developer tools

☐ **Stable internet connection** — video call + development tools running simultaneously

☐ **Free access to these websites** — GitHub.com · Vercel.com · Looperman.com · Claude.ai *(students in regions where these sites are restricted will not be able to participate)*

☐ **English reading ability** — all lessons are taught and written in English

> 💡 **Claude AI account:** No action needed. The teacher provides and manages all Claude accounts used in lessons.

🎓 Entry Requirements

Recommended Path

Complete Foundation Level first. Foundation Level builds the confidence, vocabulary, and prompt engineering instincts that make Studio Level significantly easier and more enjoyable.

Direct Entry by Evaluation

Students who have not completed Foundation Level are welcome to apply. The teacher will conduct a short conversation to confirm readiness. The evaluation covers:

Area	What We Check
Coding concepts	Can you explain what if/else does? What is a loop? Can you read simple logic and predict what it does?
AI concepts	What is AI? What is Big Data? How does a machine learn from examples?
Prompt skills	Have you used AI to create something — an image, a plan, a piece of writing? Can you write a clear, specific prompt to get a useful result?

> There are no wrong answers in the evaluation. The goal is to understand where the student is, so the teacher can give them the right support from day one.